

Probabilistic Ontologies

A Probabilistic Ontology in $\mathcal{BEL}$

x0: respiratory-virus season

$\langle \text{FluLikeCase} \sqsubseteq \exists \text{hasSymptom.Cough} : x0 \rangle$

$\langle \text{FluLikeCase} \sqsubseteq \exists \text{hasSymptom.Fever} : x0 \rangle$

$\langle \text{ClinicCase} \sqsubseteq \text{FluLike Case} : x0 \rangle$

x4: the patient belongs to a high-risk group

$\langle \text{ClinicCase} \sqsubseteq \text{HighRiskCase} : x4 \rangle$

x1: recently vaccinated

$\langle \exists \text{hasSymptom.Cough} \sqsubseteq \text{RespiratoryCase} : \neg x1 \rangle$

x2: fever observation is reliable

$\langle \exists \text{hasSymptom.Fever} \sqsubseteq \text{FebrileCase} : x2 \rangle$

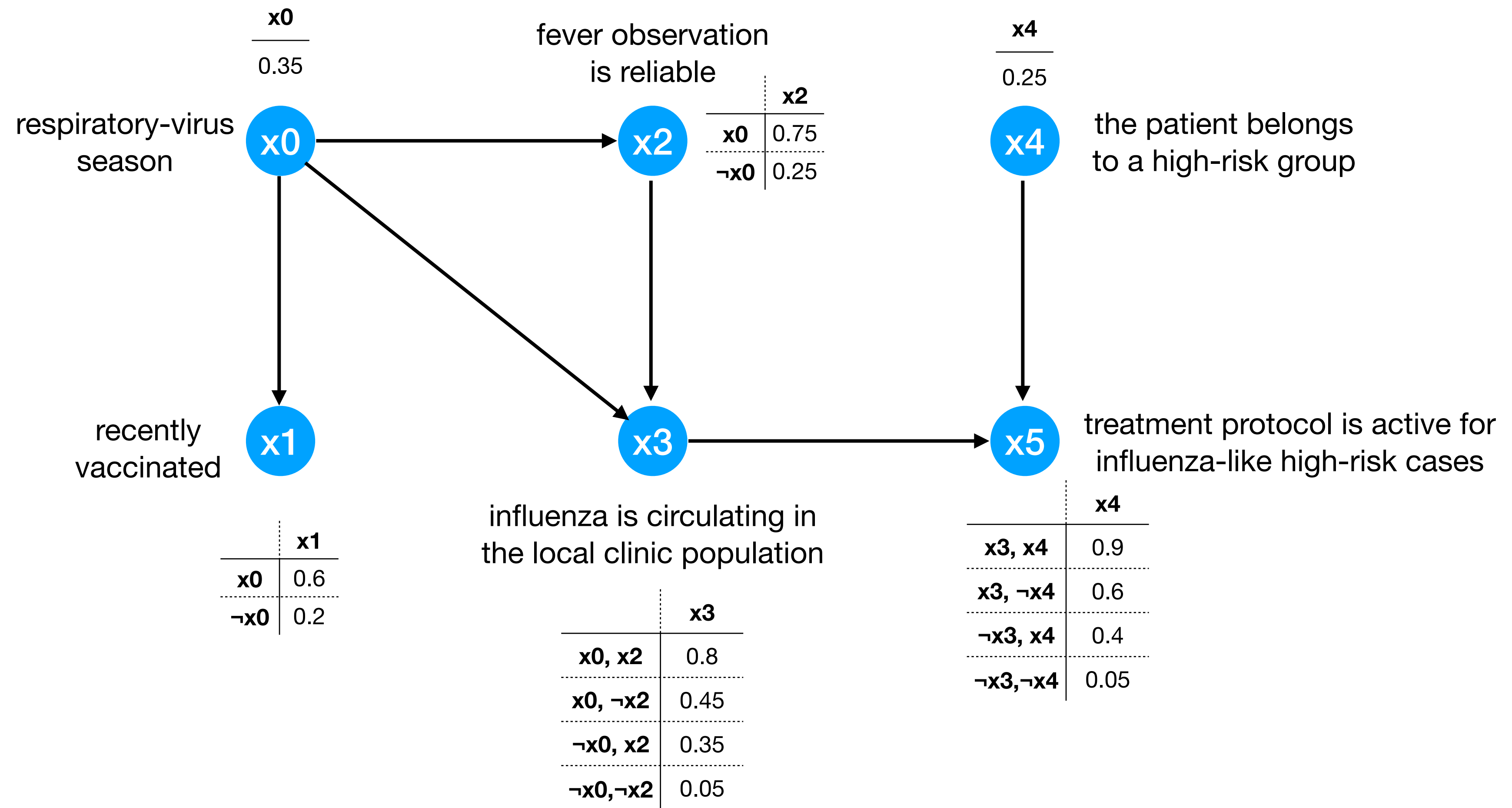
x3: influenza is circulating in the local clinic population

$\langle \text{FebrileCase} \sqcap \text{RespiratoryCase} \sqsubseteq \text{ProbableInfluenza} : x3 \rangle$

x5: treatment protocol active for influenza-like high-risk cases

$\langle \text{HighRiskPatient} \sqcap \text{ProbableInfluenza} \sqsubseteq$
 $\text{UrgentTreatmentCase} : x5 \rangle$

Bayesian Network



Derivation for

ClinicCase $\sqsubseteq$ UrgentTreatmentCase

ClinicCase

- $\sqsubseteq$ FluLikeCase [x0]
- $\sqsubseteq \exists$ hasSymptom.Fever [x0]
- $\sqsubseteq$ FebrileCase [x2]

ClinicCase

- $\sqsubseteq$ FluLikeCase [x0]
- $\sqsubseteq \exists$ hasSymptom.Cough [x0]
- $\sqsubseteq$ RespiratoryCase [$\neg$ x1]

FebrileCase $\sqcap$ RespiratoryCase

- $\sqsubseteq$ ProbableInfluenza [x3]

ClinicCase

- $\sqsubseteq$ HighRiskPatient [x4]

ProbableInfluenza $\sqcap$ HighRiskPatient

- $\sqsubseteq$ UrgentTreatmentCase [x5]

ClinicCase $\sqsubseteq$ UrgentTreatmentCase
is supported in the context:

x0, $\neg$ x1, x2, x3, x4, x5

Under the supplied Bayesian network
this context has probability:

0.0189

Example Files – Campus Virtual

- flu_example.owl OWL 2 EL ontology in functional syntax
- flu_bn.pl Bayesian network in ProbLog syntax
- flu_example_README.md README file

flu (http://example.org/born/flu) : [/Users/schorlemmer/Documents/BORN/flu_example.owl]

<>

flu (http://example.org/born/flu)

↕ 🔍

BORN × Active ontology × Entities × Individuals by class × DL Query × SPARQL Query ×

Ontology header:

⌵ ⌵ ⌵ ⌵

Ontology IRI

http://example.org/born/flu

Ontology Version IRI

e.g. http://example.org/born/flu/1.0.0

Annotations +

Ontology metrics:

⌵ ⌵ ⌵ ⌵

Metrics

Axiom	19
Logical axiom count	8
Declaration axioms count	11
Class count	9
Property count	1
ObjectProperty count	0
SymmetricObjectProperty count	0
TransitiveObjectProperty count	0
AsymmetricObjectProperty count	0
ReflexiveObjectProperty count	0
IrreflexiveObjectProperty count	0
FunctionalObjectProperty count	0
TransitiveProperty count	0
AsymmetricProperty count	0
ReflexiveProperty count	0
IrreflexiveProperty count	0
FunctionalProperty count	0
Domain	0
Range	0
Domain of ObjectProperty	0
Range of ObjectProperty	0
Domain of Property	0
Range of Property	0
Domain of TransitiveProperty	0
Range of TransitiveProperty	0
Domain of AsymmetricProperty	0
Range of AsymmetricProperty	0
Domain of ReflexiveProperty	0
Range of ReflexiveProperty	0
Domain of IrreflexiveProperty	0
Range of IrreflexiveProperty	0
Domain of FunctionalProperty	0
Range of FunctionalProperty	0

Annotations for SubClassOf

FebrileCase and RespiratoryCase SubClassOf ProbableInfluenza

Annotations +

prob

x3

⌵ ⌵ ⌵ ⌵

OK

Ontology imports Ontology Prefixes General class axioms

General class axioms

⌵ ⌵ ⌵ ⌵

General class axioms +

● FebrileCase and RespiratoryCase SubClassOf ProbableInfluenza

● HighRiskPatient and ProbableInfluenza SubClassOf UrgentTreatmentCase

● hasSymptom some Cough SubClassOf RespiratoryCase

● hasSymptom some Fever SubClassOf FebrileCase

⌵ ⌵ ⌵ ⌵

⌵ ⌵ ⌵ ⌵

⌵ ⌵ ⌵ ⌵

⌵ ⌵ ⌵ ⌵

To use the reasoner click Reasoner > Start reasoner

☒ Show Inferences

⌵

untitled-ontology-417 (http://www.semanticweb.org/schorlemmer/ontology-417) : [http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-417]

untitled-ontology-417 (http://www.semanticweb.org/schorlemmer/ontology-417)

Active ontology x BORN x Entities x Individuals by class x DL Query x SPARQL Query x

BORN: Reasoner Tests Experiments

ONTOLOGY /Users/schorlemmer/Documents/BORN/flu_example.owl

BAYESIAN NETWORK /Users/schorlemmer/Documents/BORN/flu_bn.pl

RULES

```
coni(A1),
coni(A2),
coni(B).
subx(X, exists(R, B)) :-
  gci(A, exists(R, B)),
  subx(X, A),
  coni(X),
  coni(A),
  coni(B),
  role(R).
subx(X, B) :-
  gci(exists(R, A), B),
  subx(X, exists(R, Y)),
  subx(Y, A),
  coni(X),
  coni(Y),
  coni(A),
  coni(B),
  role(R).

% Rules to avoid empty predicates of entities
con(-).
role(-).
indiv(-).
```

EXAMPLES

QUERYquery(sub('ont:ClinicCase', 'ont:UrgentTreatmentCase')).

RESULT

```
sub('ont:ClinicCase','ont:UrgentTreatmentCase'): 0.0189
```

Class hierarchy: owl:Thing

Asserted

Description: Equivalent To + SubClass Of + General class axioms +

To use the reasoner click Reasoner > Start reasoner ☒ Show Inferences